MIL-M-50023 (CmlC)

23 June 1959

(See section 6)

# MILITARY SPECIFICATION

## MASK, PROTECTIVE, CIVILIAN, CD V-805

### 1. SCOPE

1.1 This specification covers one type of civilian protective mask, made in six sizes.

### 2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

UU-P-111 - Tape; Paper, Gummed (Sealing and Securing).
UU-P-553 - Paper, Wrapping, Tissue.
UU-T-116 - Tape, Paper, Gummed, Water-Resistant.
PPP-B-636-- Boxes, Fiber.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-F-50022 - Faceblank (for CD V-805 Mask).

STANDARDS

FEDERAL

Fed. Std. No. 601 - Rubber; Sampling and Testing (Test Method).

FSC 4240

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

CHEMICAL CORPS

C5-1-288 - Mask, Protective, Civilian, CD V-805 - Assembly and Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.
(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to the specifications and drawings forming a part of this document.

3.1.1 Faceblank. The faceblank shall conform to Specification MIL-F-50022.

3.2 Manufacture and assembly. The mask shall be manufactured and assembled as specified on Drawing C5-1-288 and subsidiary drawings listed thereon.

3.3 DOP smoke penetration. DOP smoke penetration of the facepiece shall not exceed 0.03 percent when tested as specified in 4.5.1.

3.4 Outlet valve leakage. Leakage through the assembled outlet valve shall not exceed 15 milliliters of air per minute, at a pressure differential of 25 millimeters of water, when tested as specified in 4.5.3.



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3.5 Resistance to airflow.

3.5.1 Facepiece assembly. The resistance to airflow through the assembled facepeices shall not exceed the following when tested as specified in 4.5.4.1:

| | | |
|---|---|---|
| Size 1 | ---------- | 33 millimeters of water |
| Size 2 | ---------- | 31 millimeters of water |
| Size 3 | ---------- | 29 millimeters of water |
| Size 4 | ---------- | 28 millimeters of water |
| Size 5 | ---------- | 27 millimeters of water |
| Size 6 | ---------- | 26 millimeters of water |

3.5.2 Outlet valve. Resistance to airflow through the assembled outlet valve shall not exceed 15 millimeters of water when tested as specified in 4.5.4.2.

3.6 Rough handling. The values for resistance to airflow and DOP smoke penetration on the assembled facepiece after rough handling shall not increase when tested as specified in 4.5.5.

3.7 Seam and seal strength.

3.7.1 Chin seam. The average strength of the dielectric welded chin seam shall be not less than 30 pounds per inch of width, with no specimen less than 25 pounds, when tested as specified in 4.5.6.1.

3.7.2 Lens to faceblank seal. The average strength of the lens to faceblank seal shall be not less than 30 pounds, per inch of width with no specimen less than 25 pounds, when tested as specified in 4.5.6.2.

3.7.3 Buckle rivet to faceblank. The rivet shall not separate from the faceblank when tested as specified in 4.5.6.3.

3.7.4 Chin strap. The welded lap of the chin strap shall show no signs of spearation when tested as specified in 4.5.6.3.

3.7.5 Deflector seal to faceblank. The deflector shall show no signs of separation from the faceblank at the welded seals nor show any tears in deflector material when tested as specified in 4.5.6.3.

3.8 Carrier.

3.8.1 Fastener separation. The stud and washer shall be attached to the carrier in such a manner that the pressure required to open and close the fastener neither weakens nor injures the carrier when tested as specified in 4.5.7.1.



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3.8.2 Seam separation. The welded seam of the carrier shall not separate or show any signs of separation when tested as specified in 4.5.7.2.

3.9 Package leakage. The sealed bag containing the facepiece in carrier shall not leak when tested as specified in 4.5.8.

3.10 Moisture content. The average moisture content of the sample facepeices, after packaging, shall not exceed 2.0 percent and no individual facepiece shall exceed 2.5 percent moisture when tested as specified in 4.5.2.

3.11 Workmanship. The mask shall be free from contamination and damage such as permanent set of the faceblank, cracks, dents, tears, and malfunctioning of hardware.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all material and components are in accordance with 3.1 and section 5 (see 6.3).

4.3 Alternate inspection (including testing) procedures. The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 Lotting. A lot shall consist of the masks from one size faceblank, produced by one manufacturer from the same materials and under essentially the same manufacturing conditions, However, not more than one lot of faceblanks shall be represented in any one lot of finished masks.

4.4.2 Sampling.



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4.4.2.1 For nondestructive examination and tests. Sampling shall be conducted in accordance with Standard MIL-STD-105. The level and degree of inspection shall be as specified in the contract.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with the appendix to Standard MIL-STD-105 using level L7. These samples shall be tested as specified in 4.5.5, 4.5.6.1, 4.5.6.2, 4.5.7, and 4.5.2 using AQL 0.65 percent defective.

4.4.3 Examination. Sample masks shall be examined in accordance with the classification of defects and the AQL's designated by this specification and with Standard MIL-STD-105. Sample masks shall be tested as specified by 4.5.

4.4.4 Classification of defects.

4.4.4.1 Facepiece, civilian, CD V-805. (Dwg. D5-2-904 which is listed on Dwg. D5-2-903.)

| Categories and defects | | Acceptance standards |
|---|---|---|
| *Critical | | |
| A. | DOP smoke penetration | 4.5.1 |
| B. | Outlet valve leakage | 4.5.3 |
| Major A | AQL 0.15 percent defective | |
| 1a. | Assembled facepiece resistance to airflow | 4.5.4.1 |
| 2a. | Outlet valve resistance to airflow | 4.5.4.2 |
| 3a. | Package leakage | 4.5.8 |
| Major | AQL 0.65 percent defective | |
| 1. | Component missing, incorrectly located or attached | |
| 2. | Lens to faceblank seat imperfections | |
| 3. | Lens damaged, scratched, or distorted | |
| 4. | Hardware nonfunctioning | |
| 5. | Separation buckle-rivet to faceblank | 4.5.6.3 |
| 6. | Outlet valve loose in faceblank | |
| 7. | Chin seam incorrect | |
| 8. | Damage (cracks, holes, tears, or abrasions) | |
| 9. | Marking incorrect | |
| 10. | Contamination | |

*Note. Each facepiece in the production lot shall be inspected for these characteristics.



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Minor AQL 2.5 percent defective

| | | |
|---|---|---|
| 51. | Chin strap welded lap incorrect | 4.5.6.3 |
| 52. | Deflector seal to faceblank incorrect | 4.5.6.3 |
| 53. | Marking missing or illegible | |
| 54. | Coating missing | |
| 55. | Headharness stitching incorrect | |

4.4.4.2 Carrier, civilian protective mask. (Dwg. D5-4-337)

Categories and defects — Acceptance standards

Major A AQL 0.25 percent defective

| | | |
|---|---|---|
| Ia. | Open seams | 4.5.7.2 |

Major AQL 1.0 percent defective

1. Component missing, incorrectly located, or attached
2. Dimensions incorrect
3. Marking incorrect
4. Hardware nonfunctioning
5. Contamination
6. Damage

Minor AQL 2.5 percent defective

51. Coating missing
52. Marking missing, or illegible

4.4.4.3 Preparation for delivery (See section 5.)

Categories and defects

Major AQL 4.0 percent defective

1. Initial wrap missing or not secured in place
2. Packaging method incorrect or damaged
3. Containers incorrect
4. Quantity per container incorrect
5. Interlocking partitions missing
6. Closure of container incorrect
7. Damaged containers
8. Marking incorrect, missing, or illegible
9. Nonuniform quantities per pack
10. Mixed sizes of mask per pack



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Minor

None

4.5 Tests.

4.5.1 DOP smoke penetration. Prior to and after rough handling, the assembled facepiece shall be tested for DOP smoke penetration using Chemical Corps E27 smoke penetration meter.

4.5.2 Moisture content. Moisture content of the sample masks shall be determined by weighing the masks to the nearest 0.1 gram (wet wt.) immediately after unpackaging. Place the masks in a forced-air drying oven at 160° ± 5°F. and a maximum of 5 percent relative humidity until constant dry weight, within 0.1 gram, is reached. Record the final weighing (dry wt.) to the nearest 0.1 gram and determine the percentage moisture content by the following formula:

$$\text{Percent moisture content} = \frac{100 \text{ (wet wt. - dry wt.)}}{\text{wet wt.}}$$

4.5.3 Outlet valve leakage. The outlet valve assembly shall be tested for leakage using the Chemical Corps M4 Outlet Valve Leakage Indicator.

4.5.4 Resistance to airflow.

4.5.4.1 Assembled facepiece. The assembled facepiece inhalation resistance shall be determined using CmlC Inhalation and Exhalation Resistance Indicator, Q43, at a continuous airflow rate of 32 liters per minute.

4.5.4.2 Outlet valve. The outlet valve exhalation resistance shall be determined using CmlC Inhalation and Exhalation Resistance Indicator, Q43, at an airflow rate of 85 liters per minute.

4.5.5 Rough handling. After testing as specified in 4.5.1, the sample facepieces shall be placed (4 at a time) in the E4 rough handling machine and rough handled for a minimum period of 15 minutes. The facepiece shall then be retested as specified in 4.5.1 and 4.5.4.1.

4.5.6 Seam and seal strength.



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4.5.6.1 Chin seam. A rectangular specimen, 1.00 ± 0.01 inch wide and a minimum length of 2-1/2 inches, with the length perpendicular to the chin seam, shall be cut from the throat area of each sample facepiece. The width of the specimen shall be decreased to 0.75 ± 0.01 inch when necessary to avoid curvature due to the contour of the chin area. The specimen shall be placed in the testing machine with the jaws no less than 1 inch from any part of the chin seam. The test shall be conducted in accordance with method 4311 of Fed Test Method Std. No. 601 at a jaw separation rate of 12 ± 0.5 inches per minute.

4.5.6.2 Lens to faceblank seal. A rectangular specimen 1.00 ± 0.01 inch wide by 3 to 4 inches long, consisting of approximately half lens and half faceblank material with the length perpendicular to the lens-faceblank seam, shall be cut from each sample facepiece. The specimen shall be placed in the testing machine with the jaws no less than 1 inch from any part of the seal. The test shall be conducted in accordance with method 4311 of Fed. Test Method Std. No. 601 at a jaw separation rate of 12 ± 0.5 inches per minute.

4.5.6.3 Buckle-rivet, chin strap, and deflector seal. The buckle-rivet joint, chin strap seal, and deflector seal shall be thoroughly flexed and stretched by the fingers. The seals shall be inspected for adequacy, separation of the welds, and such other defects as are indicative of poor seal technique.

4.5.7 Carrier.

4.5.7.1 Fastener separation. The fasteners shall be manually operated and shall snap and unsnap readily at the point on the periphery of the socket normally used when opening and closing the fastener.

4.5.7.2 Seam separation. Seam strength tests shall be conducted in accordance with method 4111 of Federal Test Method Standard No. 601 except that the machine rate of travel shall be 2 ± 0.1 inches per minute.

4.5.7.2.1 Specimens. Die I of figure 4111 shall be used for cutting specimens and they shall be cut so that the full width of the seam will be within the constricted portion of the specimen, with the long dimension of the specimen approximately perpendicular to the seam.

4.5.8 Package leakage. Leakage of the sealed bag containing the mask and carrier shall be determined by subjecting the package to the quick-leak test in Specification MIL-P-116.



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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. With the head harness positioned inside the facepiece, each mask shall be individually wrapped in tissue paper conforming to Specification UU-P-553, securing the wrap in place with gummed tape conforming to Specification UU-T-111. Each wrapped mask shall then be placed in a plastic carrier as shown on Drawing C5-1-288, and the carrier containing the mask shall be individually packaged method IA-8 of Specification MIL-P-116.

5.1.2 Levels B and C. Not applicable.

5.2 Packing.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Forty masks of one size, packaged as specified in 5.1.1, shall be packed in one tier floored 2 by 20 in a W5c, or W6c style RSC, fiber box conforming to Specification PPP-B-636, having approximate inside dimensions of 23 3/4 by 8 by 12 1/4 inches. The box shall be divided into four equal compartments by interlocking fiberboard partitions fabricated from the same material as the box with each compartment containing ten masks. The masks shall be alternately positioned in the box so that the valve ends will be in opposite locations. The interlocking partitions shall be of the same dimensions as the inside height of the box. The box shall be closed by taping all seams including the manufacturer's joint with minimum 3-inch wide gummed or pressure sensitive tape conforming to Specification UU-T-116 or PPP-T-76 respectively, extending the tape over all corners and edges at least three inches.

5.2.3 Level C. A uniform number of masks of one size, packaged as specified in 5.1.1, shall be packed in such manner as to afford protection against damage to the masks during direct shipment from the supply source to the first receiving activity. Containers used shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers a protective mask intended for use by civilians, except children below the age of four, to protect them against chemical, bacteriological, and radiological warfare agents.



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6.2 Ordering data. Procurement documents should specify the following:

a. The title, number, and date of this specification.
b. The size or sizes required.

6.3 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 Pilot lot. A pilot lot may be required to evaluate the supplier's production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

6.5 Supersession data. This specification supersedes Chemical Corps Purchase Description No. 197-54-674, dated 29 May 1959.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:

Army - Chemical Corps

Preparing activity:

Army - Chemical Corps



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